

Exercise 1.1

(a) Let Φ be a finite set of formulas and let ψ be a formula. Then

$$\Phi \models \psi \text{ iff } \left(\bigwedge_{\varphi \in \Phi} \varphi \right) \rightarrow \psi \text{ is a tautology}$$

(b) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri.

Exercise 1.2

(a) We have $\Pi = \langle V, I, G, A \rangle$ with:

- $V = \{ \text{on}_{R,B}, \text{on}_{R,G}, \text{on}_{B,R}, \text{on}_{B,G}, \text{on}_{G,R}, \text{on}_{G,B}, \\ \text{on-table}_R, \text{on-table}_B, \text{on-table}_G, \\ \text{clear}_R, \text{clear}_B, \text{clear}_G \}$
- $I = \{ \text{on}_{G,R}, \text{on-table}_R, \text{on-table}_B, \text{clear}_G, \text{clear}_B \}$

(b) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

Exercise 1.3

This is smart:

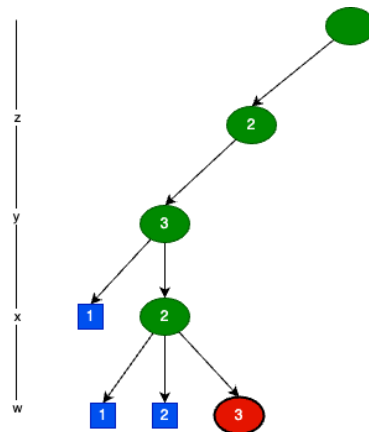


Figure 1: A very smart tree

As we can see in Figure 1 this is all very very smart.